The National School of Artificial Intelligence
Computer Architecture
Datapath Design

Assignment 05

Exercise 1:

Consider the following instruction:

Instruction: and rd, rs1, rs2

Interpretation: Reg[rd] = Reg[rs1] AND Reg[rs2]

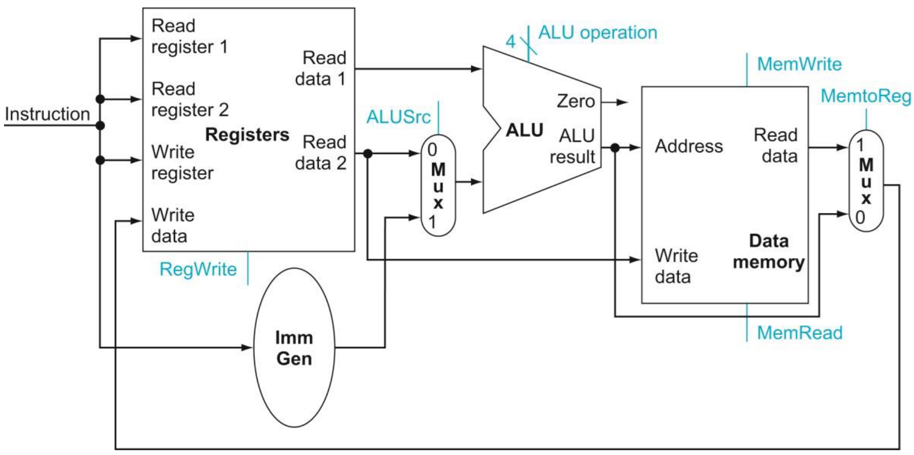

Figure 1: Exercise 1

1. What are the values of control signals generated by the control in Figure 1 for this instruction?
2. Which resources (blocks) perform a useful function for this instruction?
3. Which resources (blocks) produce no output for this instruction? Which resources produce output that is not used?
4. Explain each of the "don't cares" in the following table.

Instruction	ALUSrc	Memto-Reg	Reg-Write	Mem-Read	Mem-Write	Branch	ALUOp1	ALUOp0
R-format	0	0	1	0	0	0	1	0
lw	1	1	1	1	0	0	0	0
sw	1	X	0	0	1	0	0	0
beq	0	X	0	0	0	1	0	1

5. When silicon chips are fabricated, defects in materials (e.g., silicon) and manufacturing errors can result in defective circuits. A very common defect is for one signal wire to get "broken" and always register a logical 0. This is often called a "stuck-at-0" fault.
   a. Which instructions fail to operate correctly if the MemToReg wire is stuck at 0?
   b. Which instructions fail to operate correctly if the ALUSrc wire is stuck at 0?

Exercise 2:

Given the single-cycle CPU datapath in Figure 2

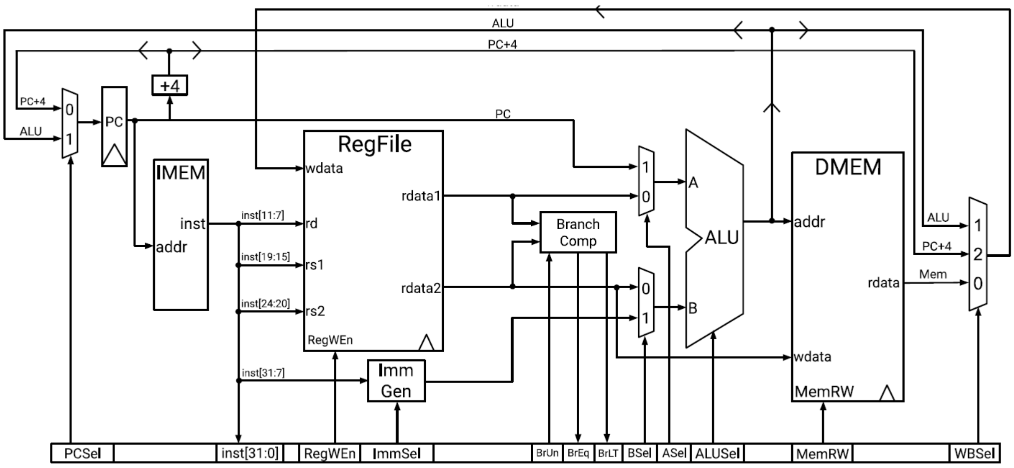

Figure 2: Single cycle datapath

1. List all possible values that each control signal may take on for the single cycle datapath, then briefly describe what each value means for each signal.
   a. PCSel
   b. RegWEn
   c. ImmSel
   d. BrEq
   e. BrLt
   f. ALUSel
   g. MemRW
   h. WBSel
2. Fill out the following table with the control signals for each instruction based on the datapath.

	BrEq	BrLT	PCSel	Imm-Sel	BrUn	ASel	BSel	ALUSel	MemRW	Reg-WEn	WB-Sel
add	*	*	0 (PC + 4)	*	*	0 (Reg)	0 (Reg)	add	0	1	1 (ALU)
ori											
lw											
sw											
beq											
jal											
blt											

Exercise 3:

Problems in this exercise assume that the logic blocks used to implement a processor's datapath (Figure 3) have the following latencies:

I-Mem / D-Mem	Register File	Mux	ALU	Adder	Single gate	Register Read	Register Setup	Sign extend	Control
250 ps	150 ps	25 ps	200 ps	150 ps	5 ps	30 ps	20 ps	50 ps	50 ps

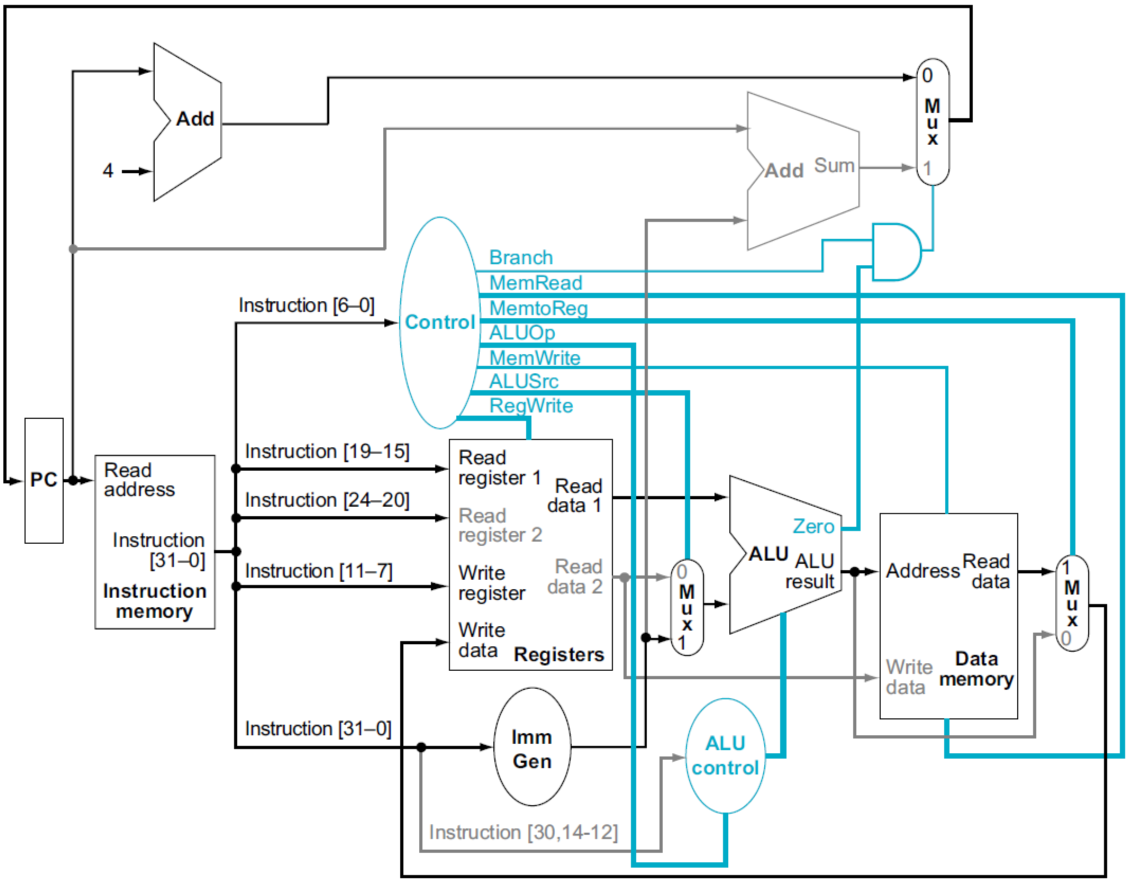

Figure 3: Datapath Exercise 3

"Register read" is the time needed after the rising clock edge for the new register value to appear on the output. This value applies to the PC only. "Register setup" is the amount of time a register's data input must be stable before the rising edge of the clock. This value applies to both the PC and Register File.

1. What is the latency of an R-type instruction, lw, sw, beq, and an arithmetic, logical, or shift I-type (non-load) instruction?
2. What is the minimum clock period for this CPU?
3. Suppose you could build a CPU where the clock cycle time was different for each instruction. What would the speedup of this new CPU be over the previous one, given the instruction mix below?

R-type/I-type (non-lw)	lw	sw	beq
52%	25%	11%	12%

4. Consider the addition of a multiplier to the CPU. This addition will add 300 ps to the latency of the ALU, but will reduce the number of instructions by 5% (because there will no longer be a need to emulate the multiply instruction).
   a. What is the clock cycle time with this improvement?
   b. What is the speedup achieved by adding this improvement?
   c. What is the slowest the new ALU can be and still result in improved performance?
5. When processor designers consider a possible improvement to the processor datapath, the decision usually depends on the cost/performance trade-off. In the following three problems, assume that we are beginning with the datapath from Figure 3, the latencies from Question 1, and the following costs:

I-Mem	Register File	Mux	ALU	Adder	D-Mem	Single Register	Sign extend	Single gate	Control
1000	200	10	100	30	2000	5	100	1	500

   Suppose doubling the number of general-purpose registers from 32 to 64 would reduce the number of lw and sw instructions executed by 12%, but increase the latency of the register file from 150 ps to 160 ps and double the cost from 200 to 400. (Use the instruction mix from Question 3.)
   a. What is the speedup achieved by adding this improvement?
   b. Compare the change in performance to the change in cost.
   c. Given the cost/performance ratios you just calculated, describe a situation where it makes sense to add more registers and describe a situation where it doesn't make sense to add more registers.

Exercise 4:

1. Examine the difficulty of adding a proposed lwi.d rd, rs1, rs2 ("Load With Increment") instruction to RISC-V.
   Interpretation: Reg[rd]=Mem[Reg[rs1]+Reg[rs2]]
   a. Which new functional blocks (if any) do we need for this instruction?
   b. Which existing functional blocks (if any) require modification?
   c. Which new data paths (if any) do we need for this instruction?
   d. What new signals do we need (if any) from the control unit to support this instruction?
2. Examine similarly the difficulty of adding a proposed swap rs1, rs2 instruction to RISC-V
   Interpretation: Reg[rs2]=Reg[rs1]; Reg[rs1]=Reg[rs2]
3. Examine the difficulty of adding a proposed ss rs1, rs2, imm (Store Sum) instruction to RISC-V.
   Interpretation: Mem[Reg[rs1]]=Reg[rs2]+immediate
4. For which instructions (if any) is the Imm Gen block on the critical path?